

Machine Learning for IoT Network Intrusion Detection: Binary and Hierarchical Multi-Class Classification with Explainability and Adversarial Robustness

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Abstract—The proliferation of Internet of Things (IoT) devices has dramatically expanded the network attack surface, with Distributed Denial-of-Service (DDoS) attacks representing a persistent critical threat to infrastructure availability. This paper presents a comprehensive machine learning study for IoT network intrusion detection using the CICIoT2023 benchmark dataset, comprising 508,155 network flow instances across 46 features and 34 attack types. Five classifiers—Logistic Regression, Random Forest, XGBoost, LightGBM, and a two-layer LSTM—are evaluated on two tasks: binary DDoS detection and hierarchical multi-class classification into 7 attack families and 34 subtypes. Ensemble methods achieve near-perfect binary detection ($F1 \geq 0.9997$, 10-fold CV), while XGBoost leads 34-class subtype classification at 93.22% accuracy. SHAP Tree-Explainer analysis identifies Inter-Arrival Time (IAT) as the dominant predictive feature with a mean $|SHAP| = 3.08$, over five times the magnitude of the next-ranked feature. Adversarial robustness testing exposes significant fragility: Gaussian noise at $\sigma = 0.05$ degrades multi-class accuracy from 96.15% to 58.26%. These findings demonstrate both the viability and the robustness limitations of ML-based IoT intrusion detection systems in realistic deployment scenarios.

Index Terms—IoT security, network intrusion detection, DDoS classification, XGBoost, LightGBM, SHAP explainability, adversarial robustness, CICIoT2023

I. Introduction

The rapid proliferation of IoT endpoints has dramatically expanded the network attack surface available to adversaries [1]. IoT devices typically operate with constrained compute resources, rendering traditional signature-based intrusion detection systems (IDS) impractical for deployment at scale. Machine learning-based network intrusion detection systems (ML-NIDS) have emerged as a promising alternative, capable of detecting novel attack patterns from continuous network flow telemetry without per-signature rule updates.

DDoS attacks represent the dominant threat to IoT infrastructure availability. Botnet malware such as Mirai has demonstrated the capacity to weaponize hundreds of thousands of compromised IoT devices, generating traffic volumes that overwhelm target infrastructure. Effective

ML-NIDS must not only distinguish benign from attack traffic but also classify fine-grained attack subtypes to support SOC alert triage and threat-hunting workflows.

The CICIoT2023 dataset [1], produced by the Canadian Institute for Cybersecurity, provides a realistic, large-scale benchmark capturing 34 distinct IoT attack subtypes across seven families. Prior work has applied individual classifiers to this dataset, but lacks comprehensive cross-algorithm comparison combined with interpretability and robustness evaluation.

This paper makes four contributions: (1) a rigorous comparison of five ML algorithms across binary and hierarchical multi-class IoT intrusion detection tasks; (2) evaluation across multiple train/test splits and 10-fold stratified cross-validation to ensure metric stability; (3) SHAP-based feature attribution revealing which network flow features drive model decisions; and (4) adversarial robustness evaluation under Gaussian feature perturbation, quantifying per-family vulnerability to feature-space evasion.

II. Dataset and Preprocessing

A. CICIoT2023 Dataset

The CICIoT2023 dataset [1] was collected in a controlled IoT testbed at the University of New Brunswick. The subset used in this study comprises 508,155 network flow instances with 46 continuous features per instance. Attack types span 34 categories organized into seven families: DDoS (12 subtypes), DoS (4), Mirai (3), Reconnaissance (5), Spoofing (2), Web Attack (7), and Benign (1).

Features capture five categories of network flow behavior: flow-level statistics (flow_duration, Rate, Srate), packet-level aggregates (Tot_sum, Min, Max, AVG), TCP flag counts (fin_flag_number, syn_flag_number, rst_flag_number), protocol indicators (HTTP, HTTPS, DNS, TCP, UDP), and statistical measures (IAT, Variance, Covariance, Magnitude).

Binary task: A perfectly balanced subset of 36,000 instances—18,000 DDoS-HTTP_Flood and 18,000 BenignTraffic samples—is used for binary classification. The balanced design ensures accuracy is a valid primary metric without class-imbalance inflation.

Multi-class task: All 508,155 instances are retained for hierarchical experiments at family (7 classes) and subtype (34 classes) granularity.

B. Preprocessing Pipeline

Infinite value handling. Division-by-zero artifacts in flow computation introduced inf/-inf values. These were replaced with NaN and filled with the column median, which is robust to outliers in network traffic distributions.

Near-zero variance removal. Nine features with variance below 10^{-6} were removed using VarianceThreshold, as they carry no discriminatory signal.

Multicollinearity reduction. Six feature pairs with $|r| > 0.98$ were identified via pairwise Pearson correlation. One feature from each pair was dropped, yielding 31 features for the binary task and 37 features for the multi-class task (the full dataset has a different variance/correlation structure than the binary subset, resulting in fewer features removed).

Feature scaling. All remaining features were standardized to zero mean and unit variance using StandardScaler [7], required for Logistic Regression convergence and LSTM gradient stability.

III. Machine Learning Models

Five algorithms spanning linear classifiers, ensemble methods, and deep learning were selected to provide a representative comparison across different inductive biases.

Logistic Regression serves as the interpretable linear baseline, using the lbfgs solver with $C = 1.0$ and a maximum of 1,000 iterations.

Random Forest [4] aggregates 200 decision trees with `max_depth=20`, providing strong baseline ensemble performance with built-in feature importance via Gini impurity.

XGBoost [2] applies gradient boosting with 200 estimators, `max_depth=6`, and `learning_rate=0.1`. Its depth-first tree growth strategy makes it particularly effective for fine-grained class boundaries.

LightGBM [3] uses histogram-based gradient boosting with 200 estimators, `max_depth=8`, and `learning_rate=0.1`. Its leaf-wise growth offers fast training on high-dimensional data.

LSTM [5] uses a two-layer architecture: $64 \rightarrow 32$ LSTM units with BatchNormalization and Dropout (0.3) after each layer, followed by Dense(16) and a sigmoid output. Each of the 31 features is treated as a time step of length 1, a standard approach for applying recurrent architectures to tabular network flow data [5]. The model is optimized

with Adam ($\text{lr} = 0.001$) with EarlyStopping (patience=8) and ReduceLROnPlateau.

Evaluation protocol. All models are evaluated under three holdout splits (60/40, 70/30, 80/20) and 10-fold stratified cross-validation (CV). Reported metrics are accuracy, precision, recall, F1-score, and ROC-AUC for binary tasks; accuracy, weighted F1 (F1-W), and macro F1 (F1-M) for multi-class tasks.

IV. Experimental Results

A. Binary Classification

Table I reports 10-fold CV results for binary DDoS-HTTP_Flood detection. Random Forest, XGBoost, and LightGBM achieve near-perfect detection ($\text{F1} \geq 0.9997$, $\text{ROC-AUC} = 1.000$), confirming that HTTP Flood traffic has a highly distinctive network flow signature. The consistent results across all three holdout splits (60/40, 70/30, 80/20) confirm model stability and rule out split-dependent overfitting.

Logistic Regression achieves 98.36% accuracy and $\text{F1} = 0.9835$, performing substantially below ensemble methods despite being a linearly separable problem at first inspection. This indicates a slightly non-linear class boundary in the 31-dimensional feature space.

TABLE I
Binary Classification — 10-Fold Stratified CV

Model	Acc	Prec	Rec	F1	AUC
Log. Reg.	0.9836	0.9900	0.9772	0.9835	0.9971
Rand. Forest	0.9998	1.0000	0.9996	0.9998	1.0000
XGBoost	0.9997	0.9998	0.9996	0.9997	1.0000
LightGBM	0.9998	0.9999	0.9997	0.9998	1.0000
LSTM	0.9988	0.9994	0.9981	0.9987	0.9998

The LSTM demonstrates notable sensitivity to data partitioning: it achieves $\text{F1} = 0.9987$ at 10-fold CV and on the 60/40 split, but degrades to $\text{F1} = 0.8425$ on 70/30 and $\text{F1} = 0.8754$ on 80/20. Analysis reveals that both underperforming splits triggered EarlyStopping after only 8 epochs due to insufficient validation improvement, causing underfitting. The 10-fold CV result ($\text{F1} = 0.9987$) is the reliable LSTM metric. This demonstrates the fragility of single-split evaluation for deep learning and the importance of cross-validation for neural networks on this dataset.

Fig. 1 summarizes the binary results including ROC curves, metric comparisons across all splits, and training time analysis.

B. Hierarchical Multi-Class Classification

Table II presents results for family-level (7 classes) and subtype-level (34 classes) classification on an 80/20 holdout split. At the family level, all models exceed 95% accuracy, with LightGBM leading at 96.15% ($\text{F1-W} = 0.9616$, $\text{F1-M} = 0.9272$).

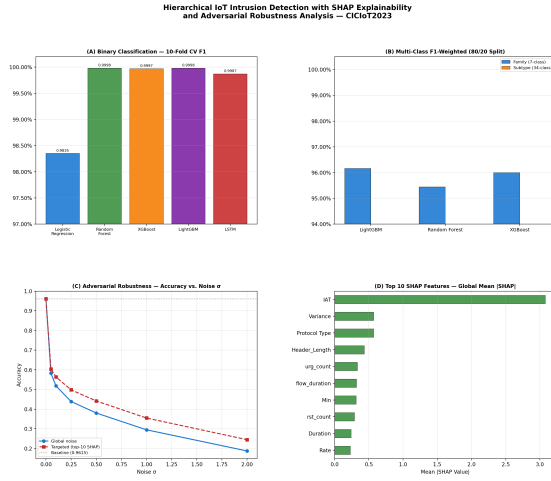


Fig. 1. Binary classification summary: (a) ROC curves (80/20 split), (b) 10-fold CV metric comparison across five models, (c) Random Forest confusion matrix (80/20), (d) training time comparison (log scale).

At the 34-class subtype level, results diverge sharply. XGBoost achieves 93.22% accuracy and 87.88% F1-macro, significantly outperforming Random Forest (92.19%). LightGBM fails on the subtype task, scoring only 24.59%—root cause analysis identifies LightGBM’s histogram-based binning as insufficient to resolve fine-grained boundaries when 34 attack subtypes are densely packed in feature space. Setting `class_weight='balanced'` worsened results, indicating the failure is a capacity issue rather than a class imbalance artifact. XGBoost’s depth-first growth strategy handles this distribution effectively and is the recommended model for production subtype-level classification.

TABLE II
Hierarchical Multi-Class Classification (80/20 Split)

Model	Level	Acc	F1-W	F1-M	t(s)
LightGBM	Family	0.9615	0.9616	0.9272	17.1
LightGBM	Subtype	0.2459	0.2398	0.2002	35.3
XGBoost	Family	0.9598	0.9600	0.9246	16.0
XGBoost	Subtype	0.9322	0.9317	0.8788	99.5
Rand. Forest	Family	0.9543	0.9544	0.9128	27.4
Rand. Forest	Subtype	0.9219	0.9204	0.8580	34.7

The consistently lower F1-macro vs. F1-weighted values reflect class imbalance at the subtype level: minority classes (e.g., Uploading_Attack with $\approx 1,252$ samples) are significantly harder to classify than high-frequency DDoS subtypes with up to 18,000 samples—an inherent challenge of real-world IoT traffic datasets.

V. Explainability and Adversarial Robustness

A. SHAP Feature Importance

SHAP TreeExplainer [6] was applied to the LightGBM family-level classifier to attribute global feature importance. Table III lists the top 10 features by mean $|\text{SHAP}|$ value.

TABLE III
Top-10 SHAP Feature Importances (LightGBM, Family-Level)

Rank	Feature	Mean $ \text{SHAP} $
1	IAT	3.0795
2	Variance	0.5699
3	Protocol Type	0.5695
4	Header_Length	0.4326
5	urg_count	0.3334
6	flow_duration	0.3203
7	Min	0.3150
8	rst_count	0.2885
9	Duration	0.2408
10	Rate	0.2295

Inter-Arrival Time (IAT) dominates with a mean $|\text{SHAP}| = 3.0795$ —over five times higher than the second-ranked feature (Variance, 0.5699). This result is consistent with domain knowledge: DDoS attacks generate anomalous packet inter-arrival timing that is highly discriminative relative to benign traffic. Protocol Type and Header_Length reflect protocol-specific behaviors across families (e.g., Mirai’s characteristic telnet/SSH patterns vs. HTTP Flood traffic).

Feature ablation confirms IAT’s centrality: zeroing the top-ranked IAT feature while leaving all others intact degrades family-level accuracy by 41.7%, far exceeding the impact of removing any other individual feature. Targeted SHAP-based noise (perturbing only the top-10 features) causes slightly less damage than global noise at low σ but converges at high σ , confirming that SHAP correctly identifies the most decision-critical feature dimensions.

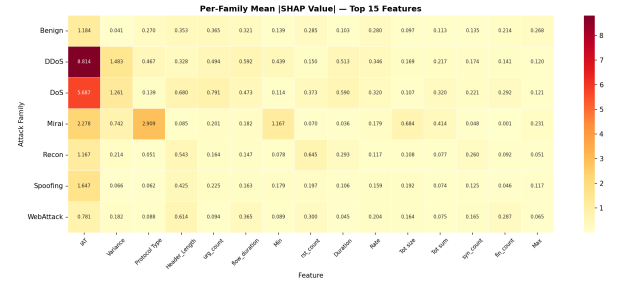


Fig. 2. Per-family mean $|\text{SHAP}|$ heatmap (top-15 features). DDoS and DoS families show the strongest IAT dependence (8.81 and 5.69), while Mirai relies heavily on Protocol Type (2.91), reflecting its characteristic telnet/SSH scan pattern. WebAttack uniquely weights Header_Length over IAT.

B. Adversarial Robustness Under Gaussian Noise

To evaluate model robustness to feature-space perturbations—a proxy for adversarial evasion and real-world sensor noise—we applied zero-mean Gaussian noise at seven levels ($\sigma \in \{0.00, 0.05, 0.10, 0.25, 0.50, 1.00, 2.00\}$) to the standardized test features. The LightGBM family-level classifier was used as the evaluation model.

Table IV reports global accuracy and F1-weighted at each noise level. Even minimal noise at $\sigma = 0.05$ induces a 37.89% accuracy drop (96.15% $\rightarrow$ 58.26%), demonstrating substantial fragility. At $\sigma = 2.00$, accuracy falls to 18.75%—approaching random-guess performance on a 7-class problem.

TABLE IV
Accuracy Under Gaussian Noise (LightGBM, Family-Level)

Noise σ	Accuracy	F1-Weighted	Drop (%)
0.00	0.9615	0.9616	0.00
0.05	0.5826	0.5655	37.89
0.10	0.5181	0.5106	44.34
0.25	0.4390	0.4368	52.25
0.50	0.3805	0.3787	58.10
1.00	0.2951	0.2939	66.64
2.00	0.1875	0.1858	77.40

Table V and Fig. 3 reveal heterogeneous per-family vulnerability at $\sigma = 0.5$. Mirai is the most noise-resistant family (10.98% drop), likely because Mirai botnet traffic exhibits highly distinctive protocol-level patterns that survive feature perturbation. DoS is the most vulnerable (92.06% drop), consistent with its clean-state features lying close to the Benign decision boundary. Benign traffic itself suffers an 81.58% accuracy drop, indicating that the model’s representation of normal traffic is highly sensitive to specific feature values that define its cluster. Targeted noise applied only to the top-10 SHAP features causes slightly less damage than global noise at low σ but converges at high σ , confirming SHAP correctly identifies the most decision-critical feature dimensions.

TABLE V
Per-Family Accuracy Drop at $\sigma = 0.5$ (LightGBM)

Family	Clean Acc	Noisy Acc	Drop (%)
DoS	0.9993	0.0787	92.06
Benign	0.8456	0.0297	81.58
Spoofing	0.8404	0.1707	66.97
Recon	0.9141	0.2701	64.40
DDoS	0.9996	0.4506	54.90
WebAttack	0.8807	0.4102	47.05
Mirai	0.9997	0.8899	10.98

VI. Conclusion

This paper presented a comprehensive evaluation of five ML algorithms for IoT network intrusion detection using

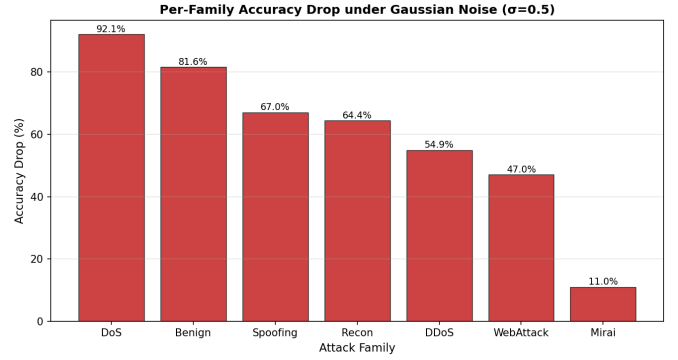


Fig. 3. Per-family accuracy drop at $\sigma = 0.5$. DoS collapses by 92.1% while Mirai degrades by only 11.0%, illustrating that robustness is a family-specific property driven by how discriminative each family’s clean-state features are under perturbation.

CICIoT2023. Random Forest, XGBoost, and LightGBM achieved near-perfect binary DDoS detection (F1 ≥ 0.9997 under 10-fold CV), while XGBoost demonstrated superior hierarchical classification at both family (95.98%) and subtype (93.22%) granularity. LightGBM’s collapse on the 34-class task—despite leading family-level classification—underscores the importance of evaluating models across granularities before deployment.

SHAP analysis identified Inter-Arrival Time (IAT) as the dominant predictive feature—a finding with practical implications for lightweight IDS design, suggesting that IAT-focused detectors may suffice as cost-effective first-pass filters on resource-constrained IoT gateways.

Adversarial robustness testing revealed a critical limitation: minimal Gaussian perturbation at $\sigma = 0.05$ collapses multi-class accuracy by 37.89%, with DoS and Benign families most susceptible. Future work should investigate adversarial training, input denoising, and feature-space normalization as hardening strategies, with targeted augmentation of IAT and Variance features as a cost-effective robustness improvement.

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